3-2 Threats and Countermeasures in Information Security [2]

Point!

1 Passwords and Authentication

(1) Password: A string of characters used to verify that the user of a given user ID is the legitimate account holder.

(2) Guidelines for creating passwords:
- Use a string that is as ([1]**long**) as possible.
- ([2]**Combine**) uppercase letters, lowercase letters, numbers, and symbols.
- Do not use personal information such as your birthday, email address, or user ID.
- Do not reuse passwords used for other services.

(3) ([3]**One-time-password**) : A password that changes at regular intervals and can only be used once.

(4) ([4]**Authentification**) : The process of verifying the identity of a user on a computer or network.

(5) Types of authentication:

Name	Method	Examples
[1] ([5]Knowledge-based authentication)	Authentication using information known only to the individual.	User ID and password, PIN code
[2] ([6]Biometric authentication (Biometrics))	Authentication using physical or behavioral characteristics of the individual.	Fingerprint, Iris, Vein pattern, Handwriting
[3] ([7]Possession-based authentication)	Authentication using an item that the individual possesses.	IC card, One-time password, SMS-based verification

[4] ([8]**Two-factor authentication**) : A method that combines two different types of factors from "knowledge," "biometrics," and "possession" to perform authentication.

[5] ([9]**Two-step authentication**) : A method that performs authentication in two steps using two pieces of information from the same type of factor.

2 Information Security Measures

(1) ([10]**Access control**) : A method of limiting access to computer systems or data so that only specific users, verified through authentication, are allowed to use them.

(2) ([11]**Firewall**) : A system installed at network entry points to prevent ([12]**unauthorized access**) from outside and to stop ([13]**data leaks**) from within.

(3) Countermeasures against computer viruses
- Install ([14]**antivirus software**) to remove or isolate viruses, and keep the virus definitions within the software up to date.
- Always keep the operating system (OS) and application software ([15]**updated**) to prevent ([16]**security holes**) (vulnerabilities) in the software.
- Regularly create ([17]**backups**) of your data.

Warm Up

Answer the following questions.

(1) Choose the one incorrect statement regarding password best practices from the options **A** to **D** below, and answer using the letter.

A Do not reuse the same password across multiple services.
B It is best to keep using the default password that was initially assigned.
C Combine letters, numbers, and symbols when creating a password.
D Avoid using easily guessable information such as your name or birthday.

(2) Choose all correct statements about one-time passwords from the options **A** to **D** below, and answer using the letters.

A If a one-time password is leaked, it can easily lead to unauthorized access.
B Using a one-time password strengthens overall security.
C A one-time password has a limited usage time and becomes invalid after expiration.
D It can prevent unauthorized access using leaked passwords.

(3) Choose the one correct example of biometric authentication from the options **A** to **D** below, and answer using the letter.

A Authentication using a user ID and password assigned to each individual
B Authentication using an SMS sent to a smartphone
C Authentication by scanning a fingerprint on a sensor
D Authentication using a one-time password

(4) If a password can use the digits 0 to 9 and lowercase letters a to z, how many different combinations are there for a 3-character password? Give your answer in the form of a^n.

Explanation

(1) If an initial password is assigned via email or memo, there is a possibility that the password has been leaked to a third party. Therefore, the initial password must be changed. The correct answer is **B**.

(2) A one-time password is a password that changes at fixed intervals and can only be used once. This strengthens security. The correct answers are **B, C,** and **D**.

(3) Biometric authentication refers to the use of an individual's physical or behavioral characteristics for verification. Examples include fingerprint, iris, vein, or handwriting-based authentication. The correct answer is **C**.
Note: A is knowledge-based authentication, B is possession-based authentication, and D is also possession-based authentication.

(4) There are 10 digits (0–9) and 26 letters (a–z), making 36 possible characters. Since each character in the password can be any of the 36, the total number of combinations for a 3-character password is 36 × 36 × 36 = **36^3 combinations**.

Try

Answer the following questions.

(1) Choose the one incorrect statement regarding password best practices from the options **A** to **D** below, and answer using the letter.

A Do not use information such as phone numbers, birthdays, email addresses, or user IDs.
B It is best to continue using the initial password without changing it.
C Do not reuse the same password across different services.
D Use a mix of uppercase and lowercase letters, numbers, and symbols.

(2) Choose the one thing that can be prevented by using a one-time password from the options **A** to **D** below, and answer using the letter.

A Password theft during transmission over a network
B Tampering with confidential files after unauthorized access
C Infection by a virus through malicious software
D Unauthorized access using a leaked password

(3) Choose the one correct example of biometric authentication from the options **A** to **D** below, and answer using the letter.

A Authentication using the shape of a fingerprint or vein pattern
B Authentication using a digital certificate
C Authentication based on whether the user can correctly read distorted text in an image
D Authentication using a one-time password

(4) Choose the terms that best fit into the blanks [1] to [4] from the options **A** to **F** below, and answer using the letters.

To protect computers and networks from threats such as unauthorized access and computer viruses, it is necessary to implement various security measures. For example, determining whether a person is authorized to access a computer or network is called ([1]). As countermeasures against computer viruses, there are the introduction of ([2]) and the ([3]) of hardware and operating systems. Furthermore, a system that hides internal LAN computers from external networks and prevents unauthorized access is called a ([4]).

A Firewall **B** Antivirus software **C** Encryption
D Authentication **E** Update **F** Security hole

(5) What is the term for restricting access so that only specific users can operate a computer system or network?

♪ (6) If a password uses 26 characters (A to Z), how many times greater is the maximum number of brute-force attempts required to crack the password when increasing the length from 4 characters to 6 characters?

Exercise

1 Answer the following questions.

(1) Choose the one incorrect statement regarding password creation from the options **A** to **D** below, and answer using the letter.

A Use the shortest possible string to make it easy to remember.
B Do not reuse passwords used for other services.
C Do not write passwords down in a notebook or on sticky notes.
D Combine uppercase and lowercase letters, numbers, and symbols.

(2) Choose the one threat that can be prevented by using a one-time password from the options **A** to **D** below, and answer using the letters.

A Theft of user ID through social engineering
B Unauthorized access through brute-force attacks
C Unauthorized access using a leaked password
D Virus infection through a security hole

(3) Choose the one correct example of biometric authentication from the options **A** to **D** below, and answer using the letter.

A Authentication using a personal ID or password
B Authentication using physical characteristics such as fingerprints or irises
C Authentication based on an individual's problem-solving ability
D Authentication using physical performance such as grip strength or flexibility

(4) Choose the terms that best fit into the blanks [1] to [5] from the options **A** to **F** below, and answer using the letters.

> System administrators must implement measures such as installing a ([1]) to prevent unauthorized access from outside the system and to minimize data tampering or leakage as much as possible. In order to deal with the constant emergence of new ([2]), it is essential to introduce ([3]) and patch ([4]).
> To prevent unauthorized access, ([5]) is useful for restricting system or network usage to specific users only.

A Firewall **B** Antivirus software **C** Security
D Access control **E** Computer virus **F** Security hole

(5) What is the name of the authentication method that combines two different elements from "knowledge," "biometric," and "possession"?

♪ (6) If a password can use the digits 0–9 and lowercase letters a–z, how many possible combinations are there for a 4-character password? Give your answer in the form of a^n.